



Selva Ganesh L 251801302-2025-AIDS ▾

S2**Started on** Thursday, 8 October 2026, 10:14 AM**State** Finished**Completed on** Thursday, 8 October 2026, 10:15 AM**Time taken** 50 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2
3  int main()
4  {
5      int n;
6      scanf("%d",&n);
7
8      int count[n+1];
9
10     for(int i=0;i<=n;i++)
11         count[i]=0;
12
13     for(int i=0;i<n;i++)
14     {
15         int x;
16         scanf("%d",&x);
17
18         count[x]++;
19
20         if(count[x]==2)
21         {
22             printf("%d",x);
23             return 0;
24         }
25     }
26
27     return 0;
28 }
```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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S2

Started on	Thursday, 8 October 2026, 10:15 AM
State	Finished
Completed on	Thursday, 8 October 2026, 10:16 AM
Time taken	34 secs
Marks	1.00/1.00

Grade 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2
3  int main()
4  {
5      int n;
6      scanf("%d",&n);
7      int a[n];
8
9      for(int i=0;i<n;i++)
10         scanf("%d",&a[i]);
11
12     for(int i=0;i<n;i++)
13     {
14         for(int j=i+1;j<n;j++)
15         {
16             if(a[i]==a[j])
17             {
18                 printf("%d",a[i]);
19                 return 0;
20             }
21         }
22     }
23
24     return 0;
25 }
```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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S2**Started on** Thursday, 8 October 2026, 10:16 AM**State** Finished**Completed on** Thursday, 8 October 2026, 10:17 AM**Time taken** 33 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2
3  int main()
4  {
5      int T;
6      scanf("%d",&T);
7
8      while(T--)
9      {
10         int n1,n2;
11         scanf("%d",&n1);
12
13         int a[n1];
14         for(int i=0;i<n1;i++)
15             scanf("%d",&a[i]);
16
17         scanf("%d",&n2);
18
19         int b[n2];
20         for(int i=0;i<n2;i++)
21             scanf("%d",&b[i]);
22

```

```
23     int i=0,j=0,first=1;
24
25     while(i<n1 && j<n2)
26     {
27         if(a[i]==b[j])
28         {
29             if(!first)
30                 printf(" ");
31             printf("%d",a[i]);
32             first=0;
33             i++;
34             j++;
35         }
36         else if(a[i]<b[j])
37             i++;
38         else
39             j++;
40     }
41     printf("\n");
42 }
43
44
45 return 0;
46 }
```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.



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S2**Started on** Thursday, 8 October 2026, 10:18 AM**State** Finished**Completed on** Thursday, 8 October 2026, 10:25 AM**Time taken** 6 mins 17 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2
3  int main()
4  {
5      int t;
6      scanf("%d",&t);
7
8      while(t--)
9      {
10         int n1,n2;
11         scanf("%d",&n1);
12
13         int a[n1];
14         for(int i=0;i<n1;i++)
15             scanf("%d",&a[i]);
16
17         scanf("%d",&n2);
18
19         int b[n2];
20         for(int i=0;i<n2;i++)
21             scanf("%d",&b[i]);
22

```

```
23     int i=0,j=0;
24
25     while(i<n1&& j<n2)
26     {
27         if(a[i]==b[j])
28         {
29             printf("%d ",a[i]);
30             i++;
31             j++;
32         }
33         else if(a[i]<b[j])
34             i++;
35         else
36             j++;
37     }
38
39     printf("\n");
40 }
41
42 return 0;
43 }
```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.



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S2**Started on** Thursday, 8 October 2026, 10:25 AM**State** Finished**Completed on** Thursday, 8 October 2026, 10:29 AM**Time taken** 4 mins 5 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2
3  int main()
4  {
5      int n,k;
6      scanf("%d",&n);
7
8      int a[n];
9
10     for(int i=0;i<n;i++)
11         scanf("%d",&a[i]);
12
13     scanf("%d",&k);
14
15     int i=0,j=1;
16
17     while(i<n&&j<n)
18     {
19         if(i==j)
20         {
21             j++;
22             continue;
23         }
24
25         if(a[j]-a[i]==k)
26         {
27             printf("1");
28             return 0;
29         }
30         else if(a[j]-a[i]<k)
31             j++;
32         else
33             i++;
34     }
35
36     printf("0");
37
38     return 0;
39 }
```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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S2**Started on** Thursday, 8 October 2026, 10:32 AM**State** Finished**Completed on** Thursday, 8 October 2026, 10:47 AM**Time taken** 15 mins 8 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2
3  int main()
4  {
5      int n,k;
6      scanf("%d",&n);
7
8      int a[n];
9
10     for(int i=0;i<n;i++)
11         scanf("%d",&a[i]);
12
13     scanf("%d",&k);
14
15     int i=0,j=1;
16
17     while(i<n&&j<n)
18     {
19         if(i==j)
20         {
21             j++;
22             continue;
23         }
24
25         if(a[j]-a[i]==k)
26         {
27             printf("1");
28             return 0;
29         }
30
31         if(a[j]-a[i]<k)
32             j++;
33         else
34             i++;
35     }
36
37     printf("0");
38
39     return 0;

```

40 | J

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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